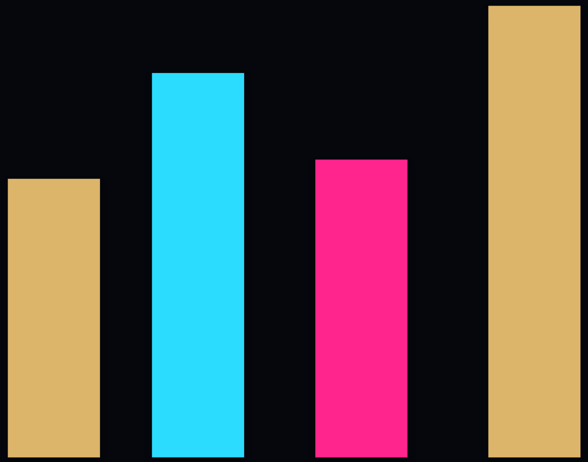
ISCARB

CPIT-455 / WEEK 01

Software Engineering (II)

Kickoff: from definitions to engineering decisions



PRIMARY SOURCE

Prof. Adeeb Noor | Fall Semester 2026/2027

"Design is how it works."

- Steve Jobs

CRISIS -> MAP -> TRADE-OFF -> EVIDENCE -> VERDICT

RULE 01 - Decide under uncertainty

Engineering starts before the terminology

A strong engineer makes a bounded decision when evidence is incomplete.

DECISION

Deploy, delay, redesign, or ask for more evidence?

UNKNOWN

What evidence could reverse the first answer?

The goal is not certainty. The goal is a defensible boundary.

TIMEBOX: 60-90 sec

YOUR TASK Name one engineering decision you expect this course to make harder - and better.

RULE 02 - Domain spine

What this course trains you to do

See the semester as a decision system, not a list of chapters.

DEPENDABILITY
Can it be trusted?

RELIABILITY
Will it keep working?

SAFETY
Can failure harm?

SECURITY
Can it resist attack?

RESILIENCE
Can it recover?

REUSE / CBSE
Can we engineer efficiently?

TIMEBOX: 60-90 sec

YOUR TASK Pick the topic you think will change your engineering judgment the most.

RULE 03 - Observable outcomes

Five outcomes you must prove

Every outcome ends in evidence, not recognition.

1. Explain dependability as trust across reliability, availability, safety, security, and resilience.
2. Trace a failure across technical and sociotechnical layers.
3. Compare alternatives and name the trade-off.
4. Build a bounded verdict with inspectable evidence.
5. Defend the verdict when a constraint changes.

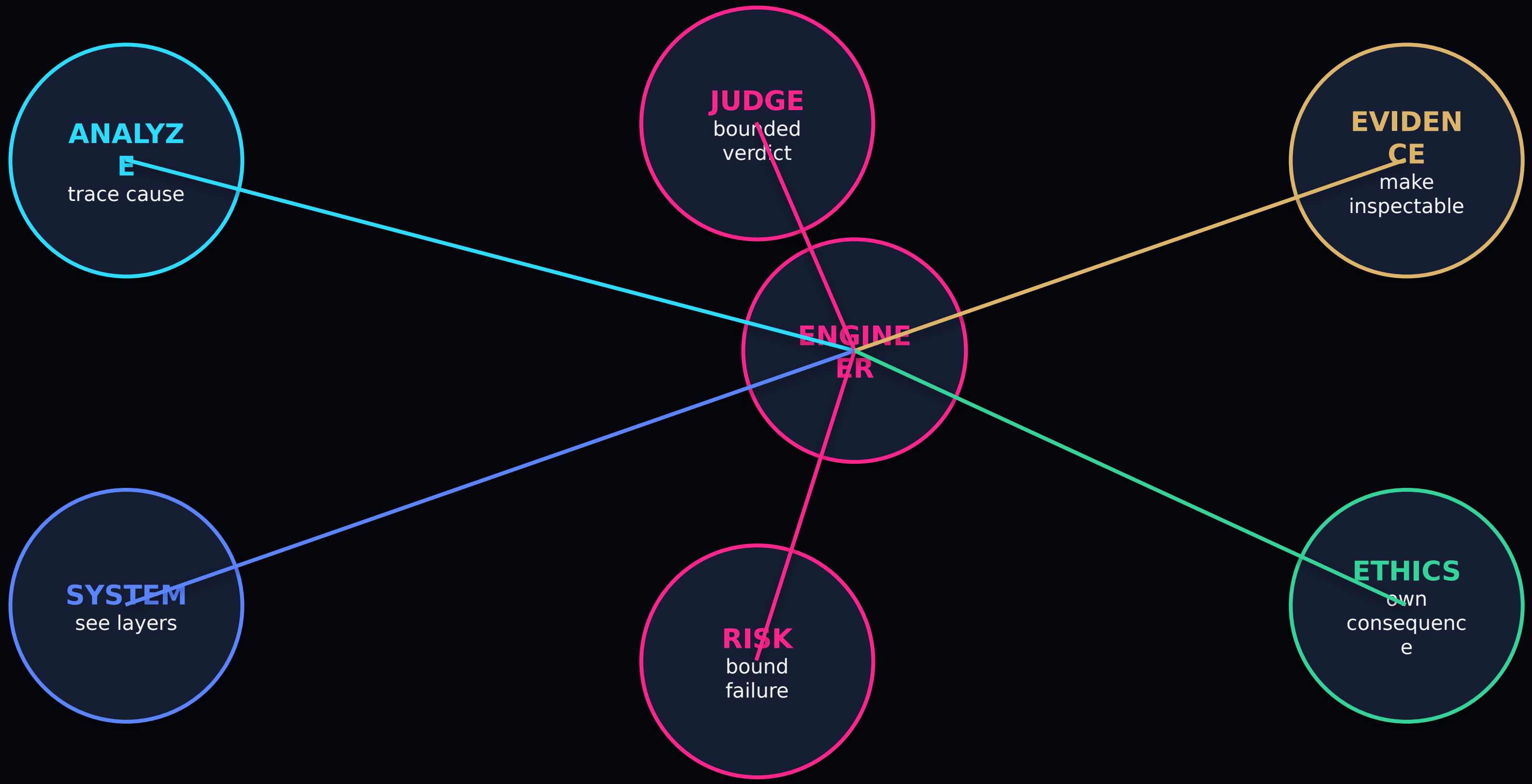
TIMEBOX: 60-90 sec

YOUR TASK Choose the outcome you expect to find hardest and state why.

RULE 04 - Engineer, not note-taker

Six engineering capabilities

Your grade follows the quality of your reasoning and evidence.



TIMEBOX: 60-90 sec

YOUR TASK Which capability fails first when a team trusts an attractive answer without evidence?

RULE 05 - Prediction gate

Do the reasoning before the reveal

Prediction forces you to expose assumptions before terminology makes the answer feel obvious.

PREDICT
What will happen?

CONSTRAINT
What limits the answer?

DERIVE
Why from first principles?

NAME
What principle did you use?

Commit first. Then compare your reasoning with the source.

TIMEBOX: 60-90 sec

YOUR TASK Make one prediction about a dependable system before seeing the source answer.

RULE 06 - ISCARB engineering flow

The five-step engineering flow

The source supplies mechanisms; the flow turns them into a decision.

1 CRISIS
What can fail?

2 MAP
Where does failure live?

3 TRADE-OFF
What do we gain/lose?

4 EVIDENCE
What can be inspected?

5 VERDICT
What is justified now?

Predict -> Constraint -> Derive -> Name

TIMEBOX: 2 min

YOUR TASK Use the flow to explain a failure you already know from software engineering.

RULE 07 - Decision card

Your minimum inspectable artifact

A good classroom answer can be audited without your oral explanation.

CLAIM
What are you saying?

EVIDENCE
What supports it?

WARRANT
Why does evidence support it?

COUNTER-EVIDENCE
What would defeat it?

UNCERTAINTY
What remains unknown?

VERDICT
Approve / revise / reject?

TIMEBOX: 2 min

YOUR TASK Write one claim and one piece of counter-evidence for the same decision.

RULE 08 - 60-second source case

ISCARB in action: Saudi hospital

The original kickoff case demonstrates the full flow in one minute.

CRISIS

Medical-device software passes all tests, but organizational sign-off is weak.

MAP

The failure is not in code; it is in the organizational layer.

TRADE-OFF

Deployment speed vs operational safety assurance.

EVIDENCE

Accountable owner + documented sign-off, not code coverage alone.

VERDICT

Reject deployment until organizational evidence is formalized.

TIMEBOX: 60-90 sec

YOUR TASK Which step in the flow actually changes the decision from deploy to reject?

RULE 09 - Falsification first

Evidence must be able to prove you wrong

An answer is stronger when you state what would make you abandon it.

EVIDENCE

What observation supports the claim?

COUNTER-EVIDENCE

What observation would reverse the verdict?

Inspectability + falsifiability = a decision worth defending.

TIMEBOX: 90 sec

YOUR TASK For the hospital case, name one piece of evidence that would allow approval.

RULE 10 - Known, unknown, monitor

Make uncertainty operational

Do not write 'we are not sure' and stop.

KNOWN
Facts already supported

UNKNOWN
Decision-sensitive gap

MONITOR
Signal that uncertainty changed

Approval = evidence state + uncertainty statement + monitoring plan.

TIMEBOX: 90 sec

YOUR TASK Turn one unknown in the hospital case into a measurable monitor or trigger.

RULE 11 - Micro-case -> local case

Scaffold first, then transfer

Use the same mechanism-evidence-boundary chain twice.

MICRO-CASE - ISCARB SCAFFOLD

Two redundant monitors share one maintenance process. One process error disables both. What assumption failed?

TRANSFER TO SOURCE CASE

Saudi hospital software passes tests, but organizational sign-off is weak. Which STS layer blocks approval?

Mechanism -> Evidence -> Decision boundary

TIMEBOX: 1 min + 5 min

YOUR TASK Solve the micro-case, then reuse the same reasoning chain on the Saudi hospital case.

RULE 12 - Owner, evidence, sign-off

A dependable answer names who owns what

No owner means the assurance claim cannot be audited.

Human sign-off is an engineering control, not administrative decoration.

TIMEBOX: 2 min

YOUR TASK Name the decision owner, evidence owner, and sign-off point in the hospital case.

RULE 13 - Breaking variable

What change would break your solution?

A robust answer exposes the variable that can reverse it.

STAFFING
fewer qualified reviewers

DEADLINE
deployment date moves forward

DATA
quality or distribution changes

REGULATION
new evidence required

If the verdict never changes, you probably have not named the real variable.

TIMEBOX: 90 sec

YOUR TASK Pick one variable and state exactly how it would reverse your current verdict.

RULE 14 - Practitioner workload

Engineering changes human work

A safe design can still fail operationally if it creates unsustainable workload.

DOCUMENTATION BURDEN

What must be produced, reviewed, and maintained?

REVIEW WORKLOAD

Who validates evidence, exceptions, and changes?

DELIVERY SPEED

What schedule pressure is the team absorbing?

Trade-offs include people, process, evidence, and time - not only code.

TIMEBOX: 90 sec

YOUR TASK Name one workload cost your preferred design creates for a real practitioner.

RULE 15 - AI assist + human sign-off

AI may prepare evidence; humans own assurance

Use AI aggressively for preparation, never for unowned approval.

AI MOMENT
assist, not assurance

AI MAY PREPARE
Draft tests, summaries, probes, candidate code, or evidence structures.

AI MUST NOT OWN
Certification, approval, risk acceptance, or final assurance.

HUMAN SIGN-OFF
A named person reviews evidence and decides accept, revise, or reject.

HUMAN-IN-THE-LOOP
If AI surfaces unclear counter-evidence: pause the verdict, escalate, and verify independently.

TIMEBOX: 90 sec

YOUR TASK State one task AI may accelerate and one decision it must never own.

RULE 16 - Grade the decision

Assessment architecture

The course rewards defended engineering decisions, not polished recitation.

30% MIDTERM
decisions under simulated constraints

30% FINAL
advanced reasoning + bounded verdicts

20% ASSIGNMENTS & LAB
ISCARB Decision Cards

20% GROUP PROJECT
live Engineering Defense

Every assessment asks: What is your claim, evidence, counter-evidence, and verdict?

TIMEBOX: 90 sec

YOUR TASK Which assessment component will require the biggest change in how you study?

RULE 17 - Constraint mutation

The live Engineering Defense

You will not only present a report; the constraint will change live.

1 PRESENT
state architecture + evidence

2 MUTATE
constraint changes live

3 REASON
re-run mechanism

4 DEFEND
show changed evidence

5 VERDICT
revise only if needed

A strong design either survives mutation - or explains exactly why it changes.

TIMEBOX: 2 min

YOUR TASK Name one constraint a reviewer could mutate during your project defense.

X01 - Roadmap: Sep -> Oct

Roadmap and milestones - phase 1

These dates are taken directly from the kickoff source.

Sep 6
Kick-off & Dependability

Sep 13
Reliability

Sep 20
No Class

Sep 27
Safety

Oct 4
Security

Oct 11
Resilience

TIMEBOX: 60-90 sec

YOUR TASK Identify which phase-1 topic is most likely to change your project architecture.

X02 - Roadmap: Oct -> Dec

Roadmap and milestones - phase 2

The second half shifts from assurance to reuse, components, distribution, and defense.

Oct 18
Midterm Exam & Project Review

Oct 25
Software Reuse

Nov 1
Component-Based Engineering

Nov 8
Distributed Engineering

Nov 15
System of Systems (SoS)

Dec 20
Final Exam & Project Defense

TIMEBOX: 60-90 sec

YOUR TASK Choose the milestone that should trigger the earliest project preparation.

RULE 18 - Make the system operable

Logistics and next steps

A strong course launch removes operational ambiguity.

THEORY

Sun, Tue, Thu | 09:00-09:50 | Bldg A80, H1-10

LAB

Thu | 15:00-16:40 | Bldg B80, L0-14

CONTACT

arnoor@kau.edu.sa | Office: Bldg 31, Room 123

TODAY

Review syllabus on Blackboard and configure the lab environment.

TIMEBOX: 5 min

YOUR TASK Before leaving today, confirm Blackboard access and your lab environment.

RULE 19 - Quick card

Peer-review quick card

Use two questions in class; keep formal scoring outside the live slide.

1 IS THE EVIDENCE INDEPENDENTLY INSPECTABLE?

Could another team verify the claim without trusting your explanation or the AI/tool that prepared it?

2 WHAT WOULD MAKE THE CLAIM FAIL?

Name the variable, counter-example, missing evidence, or failure condition that would reverse the verdict.

FAST PEER REVIEW

Inspectability + falsifiability before formal scoring.

TIMEBOX: 3 min

YOUR TASK Exchange one decision card with a peer; answer Q1 and Q2 only.

RULE 20 - Take-home decision

Are you ready to engineer?

Readiness means you can make, test, and revise a bounded decision.

READINESS CHECK
next test / monitor

DECIDE
state a bounded answer

EVIDENCE
show what supports it

BREAK
name what would reverse it

OWN
name who signs off

THE GOLDEN RULE
A decision is not ready until evidence, counter-evidence, uncertainty, and ownership are visible.

TIMEBOX: 2 min

YOUR TASK Write one sentence: 'In this course, engineering means ...' using evidence and ownership.

Ready to engineer?

The course starts when the answer becomes inspectable.

CRISIS

MAP

TRADE-OFF

EVIDENCE

VERDICT

Bring the source. Make the decision. Show the evidence. Own the consequence.